Experiment 39: Customer Clustering and Visualization

Aim: To group customers using K-Means and visualize the clusters using a scatter plot.

Algorithm:

1. Import K-Means and Matplotlib.
2. Create customer spending data.
3. Apply K-Means clustering.
4. Assign customers to different groups.
5. Visualize the clusters using a scatter plot.

Code:

```
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

X = [[100, 2], [150, 3], [200, 4],
    [500, 8], [550, 9], [600, 10],
    [900, 15], [950, 16], [1000, 18]]

model = KMeans(n_clusters=3, random_state=42, n_init=10)
labels = model.fit_predict(X)

print("Customer Clusters:", labels)

plt.scatter(
   [x[0] for x in X],
   [x[1] for x in X],
   c=labels
)

plt.xlabel("Amount Spent")
plt.ylabel("Items Purchased")
plt.title("Customer Segmentation")
```

plt.show()

Input:

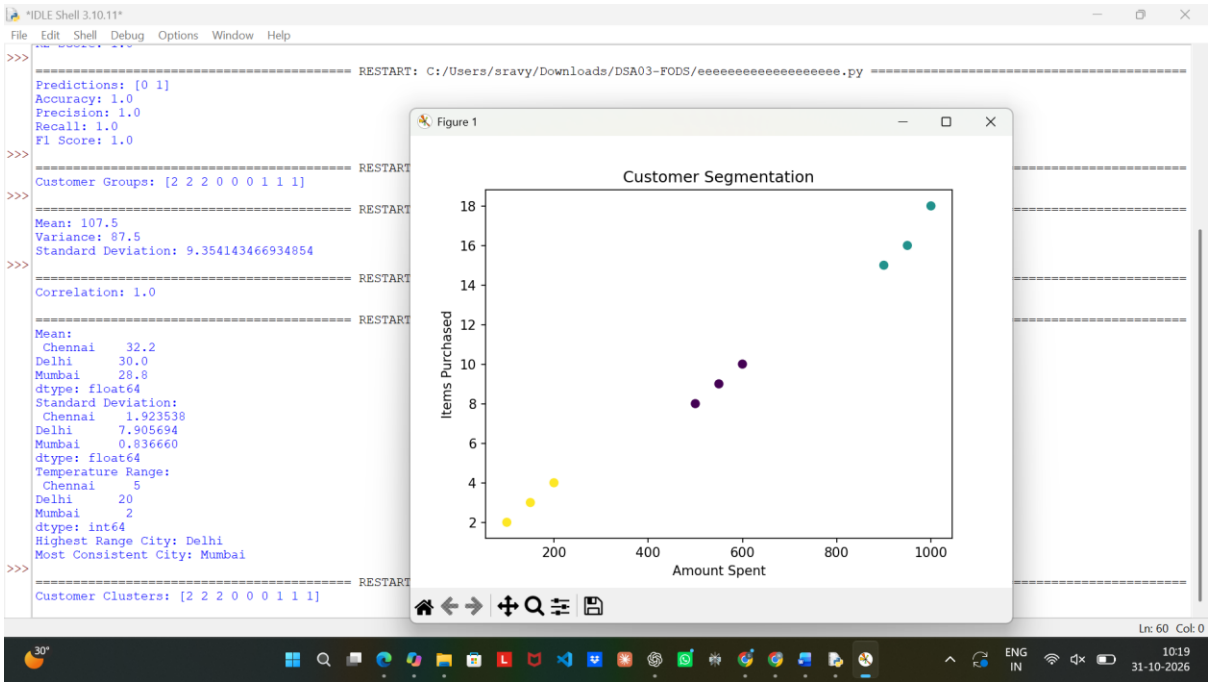
Spending: 100, 150, 200, 500, 550, 600, 900, 950, 1000

Items: 2, 3, 4, 8, 9, 10, 15, 16, 18

Output:

Customer Clusters: [1 1 1 0 0 0 2 2 2]

A scatter plot showing three customer groups will appear.



Result: Customers were grouped and visualized using K-Means clustering.